



ASTRAQUANTUM
TECH

Week 7 — SOC / Blue Team Assessment

Logging • Monitoring • Log Analysis • Alert Investigation • Incident Response

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Submission links

GitHub repository: [Insert GitHub repository link]

LinkedIn post: [Insert LinkedIn post link]

Executive Summary

This assessment documents practical work completed in authorized training environments for Week 7 of the AstraQuantum Tech Summer of Cybersecurity 2026 program. The work covered SOC fundamentals, Level 1 alert triage, log-based reasoning, incident-response decisions, and a defensive Blue Team Labs Online challenge.

The submitted evidence shows completion of the **TryHackMe SOC Fundamentals** room, completion of the **TryHackMe SOC L1 Alert Triage** room, and completion of the BTLO challenge **The Report II**. During triage, I prioritised alerts using severity and age, reviewed the surrounding context, and selected **Double-Extension File Creation** as the primary case for deeper investigation. The available evidence supports a **High-severity True Positive** verdict

because the alert describes a file named `cats2025.mp4.exe`, downloaded from an untrusted domain, with a double extension designed to make an executable appear to be a video file.

The assessment also demonstrates a clear distinction between a raw event, an alert, and an incident. A single event is not treated as proof of compromise. Instead, the verdict is based on the alert description, file naming pattern, source domain, hash value, host context, process/user context, and the recommended follow-up actions.

1. Scope and Authorized Environment

All activity described in this report was performed inside the training platforms specified by the assignment. No real organization, production endpoint, or public system was investigated. The evidence set supplied for review contains browser screenshots from TryHackMe and Blue Team Labs Online, together with the assignment brief.

The report covers three practical activities:

1. **TryHackMe — SOC Fundamentals**, including SOC roles, People/Process/Technology, detection and response, alert handling, and a port-scanning practical exercise.
2. **TryHackMe — SOC L1 Alert Triage**, including event-to-alert reasoning, severity and age-based prioritisation, alert ownership, investigation, and verdict handling.
3. **Blue Team Labs Online — The Report II**, a defensive challenge focused on SOC structure, response options, organisational models, situational awareness, and operational capabilities.

2. Tools and Platforms Used

Platform or tool	Use in this assessment
TryHackMe	SOC Fundamentals and SOC L1 Alert Triage practical rooms
Blue Team Labs Online	The Report II defensive challenge
Browser-based lab interfaces	Review of alerts, task answers, status, and completion evidence
SIEM-style alert interface	Alert fields, severity, status, assignee, verdict, host, process, target file, and hash review
Screenshot evidence	Verification of student activity, lab context, results, and system date/time

3. Practical Activity Summaries

3.1 TryHackMe SOC Fundamentals

The SOC Fundamentals room introduced the operational purpose of a Security Operations Center and the responsibilities of an L1 analyst. The practical work connected SOC roles with the People, Process, and Technology model. It also demonstrated how an analyst moves from detection to alert review, investigation, reporting, and closure or escalation.

The practical alert exercise focused on a port-scanning alert. The alert list showed “**Port Scanning Activity Detected from IP: 10.0.0.8**”, marked as a network and high-severity alert. The room completion screen recorded **7 completed tasks and 128 points**. The exercise reinforced that an L1 analyst must identify what happened, when it happened, where it occurred, who was involved, and why the activity was suspicious before making a decision.

L1 analyst summary: An L1 SOC analyst monitors alerts, validates whether an alert is meaningful, collects the surrounding evidence, records the investigation clearly, and assigns an initial verdict. The analyst should not close an alert merely because one log line appears harmless. The analyst should correlate time, source, destination, account, process, and related events. If the evidence indicates a likely attack, material impact, or insufficient information for a safe closure, the alert should be escalated with a concise evidence-based handoff. If the evidence demonstrates expected activity or a known benign explanation, the alert can be closed as benign or false positive with the reasoning documented.

Five Ws for the port-scanning alert:

- **What:** Network port-scanning activity was detected.
- **When:** The lab alert record displayed the event as **12 June 2024 at 17:24**.
- **Where:** The source shown in the alert was **10.0.0.8**, within the lab network context.
- **Who:** The visible alert identified the source IP rather than a confirmed human user. The responsible process or account would require additional host and network telemetry.
- **Why:** A port scan probes multiple services and can be used for discovery before exploitation. It is suspicious in the absence of an approved scanning purpose, but the alert alone does not prove that a compromise occurred.

3.2 TryHackMe SOC L1 Alert Triage

The SOC L1 Alert Triage room focused on the workflow from **event** → **log** → **alert** → **investigation** → **verdict**. I used the room’s prioritisation logic by reviewing severity and alert age before selecting an alert for investigation. The evidence shows the correct prioritisation responses and the selection of **Potential Data Exfiltration** as the first-priority alert. The room completion evidence records **6 completed tasks and 80 points**.

The detailed alert review also covered a **Double-Extension File Creation** alert dated **21 March 2025 at 13:58**. The record showed **High** severity, a closed status after review, and a **True Positive** verdict. The alert description explained that files such as `*.pdf.exe` or `*.gif.lnk` may be used to disguise executables and trick a user into opening them.

3.3 Blue Team Labs Online — The Report II

The selected BTLO challenge was **The Report II**, a medium-difficulty defensive challenge worth 20 points. Its scenario asked the analyst to study a MITRE report and recommend useful outcomes for a newly established SOC. The completed evidence shows correct answers covering the short-form teams **NOC, SOC, ICM**; the response options **Block Activity, Deactivate Account, Continue Watching, Refer to Outside Party**; the **OODA Loop**; a distributed SOC organisational model; the SOC Operations Lead role; the optional capabilities **Deception** and **Insider Threat**; and the virtual console technologies **iLO** and **iDRAC**.

The challenge objective was to connect SOC design with operational readiness. The key learning outcome was that an effective SOC requires defined responsibilities, repeatable workflows, situational awareness, measurable operations, and response options that are appropriate to the evidence and impact.

4. Alert Investigation Case Report

Case identification

Field	Finding
Case ID / Alert ID	TryHackMe SOC L1 Alert Triage — Double-Extension File Creation, 21 March 2025 13:58
Alert name	Double-Extension File Creation
Severity	High
Detection source	SIEM-style file-creation detection rule in the TryHackMe training environment
Status	Closed after investigation in the lab
Verdict	True Positive

Executive summary

A high-severity alert identified the creation or download of a double-extension file named **cats2025.mp4.exe**. The file was associated with the untrusted domain **freecatvideoshd.monster** and the visible record included the MD5 value **14d8486f3f63875ef93cfd240c5dc10b**. The **.mp4.exe** naming pattern is a classic attempt to make an executable look like a video file. The untrusted source and the executable extension increase the likelihood that the file is malicious or part of a phishing delivery chain. The correct lab verdict was **True Positive**, with containment and endpoint investigation recommended.

5 Ws analysis

Question	Analysis
What happened?	A suspicious double-extension executable, <code>cats2025.mp4.exe</code> , was downloaded or created.
When did it happen?	The alert record displayed 21 March 2025 at 13:58 .
Where did it happen?	The event occurred in the training environment on the host represented by the alert record. The supplied screenshot does not expose a readable host name, so the host should be obtained from the underlying SIEM record before a real containment decision.
Who was involved?	The immediate entities were the endpoint user/process that created or downloaded the file and the external domain <code>freecatvideoshd.monster</code> . The exact user and process should be confirmed from endpoint telemetry.
Why is it suspicious?	The filename uses a misleading media extension followed by <code>.exe</code> , and the source domain is untrusted. This combination is consistent with a phishing or malware-delivery technique.
Which logs support the conclusion?	The SIEM alert record, file-creation telemetry, download/proxy history, endpoint process history, DNS/network logs, and hash reputation data should be correlated. The screenshot directly confirms the alert description, URL, file name, and MD5 value.
What happened immediately before and after?	Before the alert, a user or process likely accessed the external domain and initiated a download. After the alert, the analyst reviewed the record, assigned a True Positive verdict, and closed the lab alert. In a production investigation, process execution, persistence, additional downloads, and outbound connections would be checked next.

Evidence and correlation

The most important evidence is the combination of the file name and source context. A file named `cats2025.mp4.exe` is not simply a media file; the final executable extension controls how the operating system may treat it. The domain shown in the record is not an established trusted software distribution source. The MD5 value provides a correlation key for endpoint

search and threat-intelligence lookup, although a hash alone should not be treated as proof of maliciousness.

The investigation should therefore correlate the alert with the endpoint's process tree, the user who initiated the download, browser history, DNS resolution, proxy logs, antivirus or EDR detections, and any child process created after the file appeared. This approach follows the assessment principle of evidence before assumptions.

Verdict and severity rationale

Verdict: True Positive. The lab evidence is sufficient to classify the detection as a genuine suspicious file event rather than a false positive. **Severity: High.** The file is an executable disguised as a video and was sourced from an untrusted domain. The severity would be raised further if the file executed, established persistence, contacted a command-and-control endpoint, or affected a privileged account.

Impact assessment

The immediate potential impact is endpoint compromise through user execution of a disguised executable. Possible consequences include credential theft, malware installation, persistence, lateral movement, data access, or further payload retrieval. The supplied evidence does not prove execution or successful compromise. Therefore, the confirmed finding is a high-risk malicious-looking file event, while the scope of any resulting incident remains subject to endpoint and network correlation.

Recommended response

The affected endpoint should be isolated if the file executed or if additional telemetry indicates compromise. The file should be quarantined, and the hash should be searched across endpoints. Analysts should collect the process tree, user identity, browser and proxy history, DNS records, EDR events, persistence locations, and outbound network connections. The source domain should be blocked through the appropriate secure web gateway or DNS control after confirming that it is not required for an approved business purpose. Credentials should be reset only when evidence suggests that the user or endpoint was exposed.

The case should be escalated to L2 or incident response when execution is confirmed, when the endpoint shows persistence or lateral movement, when a privileged account is involved, or when the evidence is incomplete but the potential impact is material. If the file was blocked before execution and no related activity is found, the case can be closed with the evidence and prevention action documented.

5. Alert Triage Workflow

The working triage flow used for the assessment was:

New → Assigned → Investigating → Verdict → Close / Escalate

At the investigation stage, the analyst checks the alert name, timestamp, severity, status, assignee, host, account, process, target file, source or destination, related events, and available hash or URL data. A verdict is then selected only after the evidence is compared with expected activity and surrounding events.

6. Incident Timeline

Time	Event	Evidence source	Why it matters
21 Mar 2025, 11:53	A separate Bruteforce Attack from External alert appears in the triage records.	Alert table screenshot	Establishes surrounding alert activity and the need to review the queue chronologically.
21 Mar 2025, 12:40	Unusual VPN Login Location is recorded and later marked False Positive.	Alert table and task-answer screenshot	Demonstrates that an alert can be suspicious-looking but benign after contextual review.
21 Mar 2025, 13:02	A Download alert appears in the queue.	Alert table screenshot	Provides a nearby event that could be correlated with later file activity.
21 Mar 2025, 13:30	Potential Data Exfiltration is selected as the first-priority alert in the room workflow.	Alert-prioritisation screenshot	Shows severity-and-age-based triage rather than opening alerts randomly.
21 Mar 2025, 13:58	Double-Extension File Creation is recorded as High severity.	Detailed alert screenshot	Identifies the primary case and the suspicious <code>cats2025.mp4.exe</code> file.
21 Mar 2025, after review	The alert is assigned to the L1 analyst, investigated, and marked True Positive / Closed in the training interface.	Alert detail and completion evidence	Records the analyst decision and demonstrates the full triage lifecycle.

The first five timestamps are displayed in the triage queue. The final row represents the documented analyst workflow and verdict shown in the supplied evidence; it is not presented as a separate raw endpoint log event.

7. Evidence and Screenshots

The following exhibits are selected from the submitted screenshots. They show the training platform, the student's own results, and, where visible, the system date and time. Screenshots from walkthroughs or third-party material were not used as completion evidence.

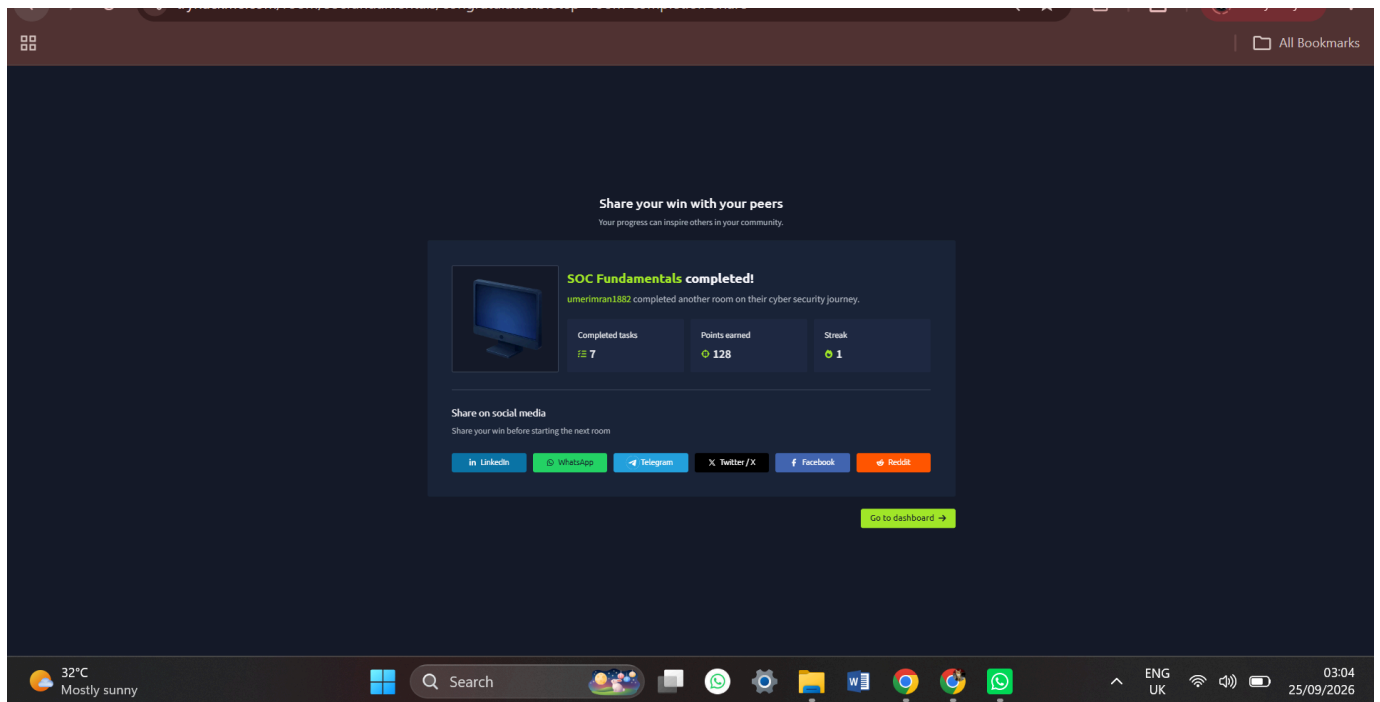
Exhibit 1 — SOC Fundamentals alert record

The screenshot displays a SIEM Alerts interface. At the top, there are buttons for 'Create alert' and 'Create mass notification'. Below these is a search bar labeled 'Search Alerts' and a toggle for 'See all alerts'. On the left, a sidebar shows 'Saved searches' with options like 'All', 'Open', 'Closed', 'Un/Acked', 'Not seen', 'Assigned to me', and 'Encrypted Alerts'. The main area lists alerts with details such as ID, severity, category, and status. A tooltip overlay indicates the flag 'THM{000_INTRO_TO_SOC}'.

Alert ID	Severity	Category	Description	Status	Resolved	Timestamp
#167	P2	Network	Port Scanning Activity Detected from IP: 10.0.0.8	CLOSED	RESOLVED	June 12, 2024 17:24
#166	P3	Network	Trojan Detected on Host: KAMRON	CLOSED	RESOLVED	June 12, 2024 13:48
#165	P4	Host	Malware	CLOSED	RESOLVED	June 12, 2024 08:26
#164	P2	Host	Critical Service Disabled on Host: GEORGE PC	CLOSED	RESOLVED	June 11, 2024 19:07

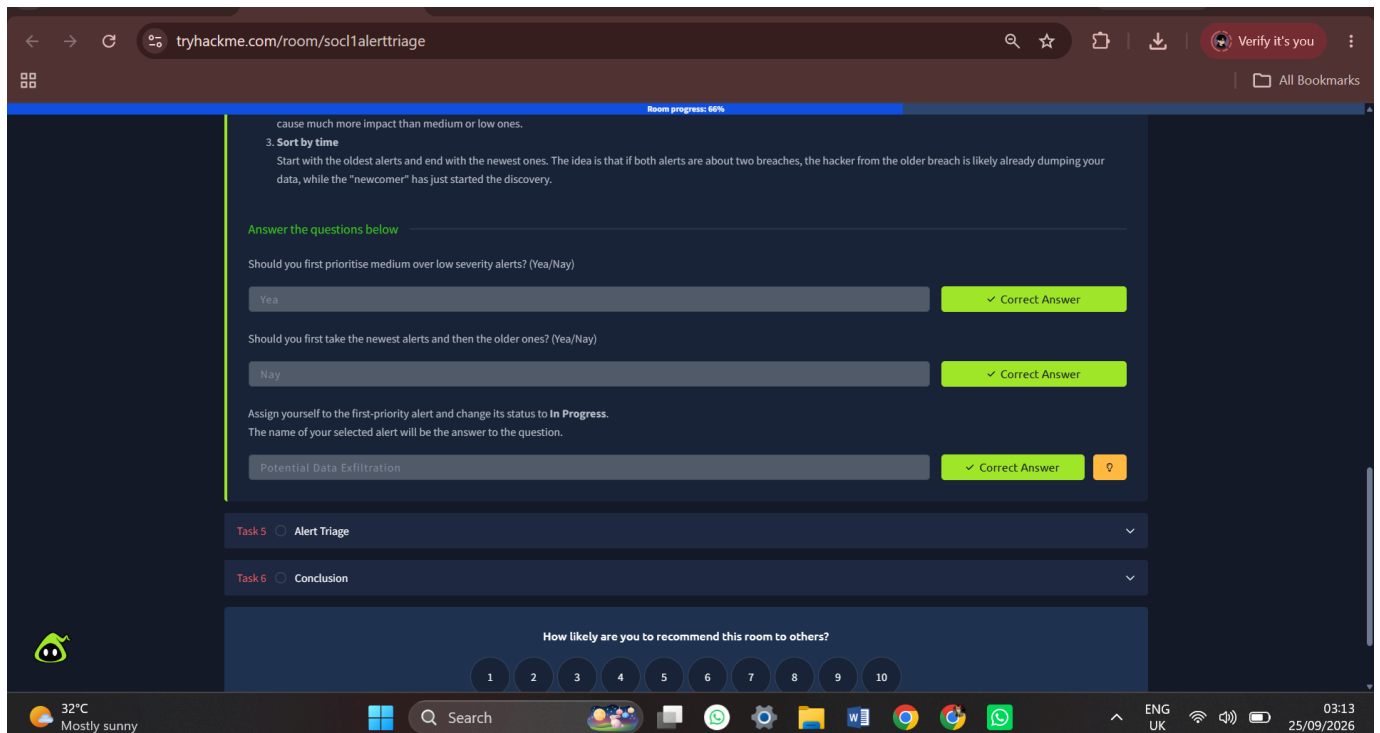
Caption: The SIEM-style lab interface shows the port-scanning alert from `10.0.0.8`, with high severity and a resolved/closed state. The screenshot also shows the lab flag generated after the student completed the exercise.

Exhibit 2 — SOC Fundamentals completion



Caption: TryHackMe completion evidence for SOC Fundamentals. The page records seven completed tasks and 128 points. The system date/time is visible at the bottom of the screenshot.

Exhibit 3 — Alert prioritisation



Caption: The SOC L1 Alert Triage room confirms the prioritisation logic and the selection of Potential Data Exfiltration as the first-priority alert. The answers are marked correct.

Exhibit 4 — Double-extension case details

static-labs.tryhackme.cloud/apps/soc1-alerttriage/

Time

Name

Severity

Status

Verdict

Assignee

Mar 21st 2025 at 13:58

Double-Extension File Creation

High

Closed

True Positive

You (L1)

Description:

Host:

Process Name:

Process User:

Target File:

File MotW:

File MD5:

Comment:

This rule detects a creation of a double-extension file like '*.pdf.exe' or '*.gif.lnk', often used by hackers in phishing attacks to trick users into opening the malicious executable.

Great work and correct verdict! Claim your flag!

THM{how_could_this_user_fall_for_it?}

https://freecatvideoshd.monster/cats2025.mp4.exe

14d8486f3f63875ef93cfd240c5dc10b

File "cats2025.mp4.exe" downloaded from untrusted domain (freecatvideoshd.monster) exhibits classic double-extension phishing technique disguising an executable as a video file. High risk of malware; genuine incident.

Caption: The detailed alert record shows High severity, the True Positive verdict, the suspicious `cats2025.mp4.exe` target, the untrusted domain, and the MD5 value used for further correlation.

Exhibit 5 — SOC L1 Alert Triage completion

Share your win with your peers

Your progress can inspire others in your community.

START

CLOSE

Triage Guide

SOC L1 Alert Triage completed!

umerimran1882 completed another room on their cyber security journey.

Completed tasks

6

Points earned

80

Streak

1

Share on social media

Share your win before starting the next room

in LinkedIn

WhatsApp

Telegram

X Twitter / X

Facebook

Reddit

Go to dashboard

Start the next room

Caption: TryHackMe completion evidence for SOC L1 Alert Triage. The page records six completed tasks and 80 points.

Exhibit 6 — BTLO challenge scenario

The screenshot shows the Blue Team Labs Online (BTLO) website interface. The browser address bar displays the URL: `blueteamlabs.online/home/challenge/the-report-ii-82ea7781c5`. The page features a dark blue header with navigation links: Home, Investigations, Challenges, Leaderboard, BECOME PRO!, and Collectibles. The main content area is divided into two columns. The left column, titled 'The Report II', provides a description of the challenge as an extension of 'The Report' challenge, aimed at SOC improvement. It includes a 'PDF Reader' button, a table of statistics (Points: 20, Difficulty: Medium, Solves: 3816, OS: Windows/Linux), a download link for 'The Report II.zip' (14MB), and a 'Download File' button. The right column, titled 'Scenario', contains a detailed description of the challenge and a 'Challenge Submission' section with two questions. Question 1 asks for the names of units/teams responsible for maintaining network and other IT equipment, incident detection and response, and security compliance and risk measurement. Question 2 asks for the 4 suggested 'Response Options' mentioned in Basic SOC Workflow. Both questions have input fields and 'Submit' buttons. The bottom of the page shows a Windows taskbar with the date 25/09/2026 and time 03:44.

blueteamlabs.online/home/challenge/the-report-ii-82ea7781c5

Home Investigations **PRO** **FREE** Challenges **FREE** Leaderboard **BECOME PRO!** Collectibles

The Report II

This challenge is an extension for an existing 'The Report' challenge where you are working in a newly established SOC where there is still a lot of work to do to make it a fully functional one. As part of the SOC improvement process, you were assigned a task to study a report released by MITRE and suggest some useful outcomes for your SOC.

[PDF Reader](#)

Points	Difficulty	Solves	OS
20	Medium	3816	Windows/Linux

[The Report II.zip](#) 14MB Password: BTLO [Download File](#)

First-Blood **Created By**

LonerVamp 1553 days ago **BTLO** 1553 days ago

Recent Solves

Scenario

This challenge is an extension for an existing 'The Report' challenge where you are working in a newly established SOC where there is still a lot of work to do to make it a fully functional one. As part of the SOC improvement process, you were assigned a task to study a report released by MITRE and suggest some useful outcomes for your SOC. Note: Answer the questions with the answers as the way you see in the document to avoid formatting issues. Report Link: <https://www.mitre.org/sites/default/files/publications/11-strategies-of-a-world-class-cybersecurity-operations-center.pdf>

Found an issue with this challenge? Click here to report it!

Challenge Submission

Question 1) Submit the name of the units/teams (in short form) that are responsible for maintaining network and other IT equipment, incident detection and response, and security compliance and risk measurement (Format: Team1, Team2, Team3) (1 points)

Format: Team1, Team2, Team3 [Submit](#)

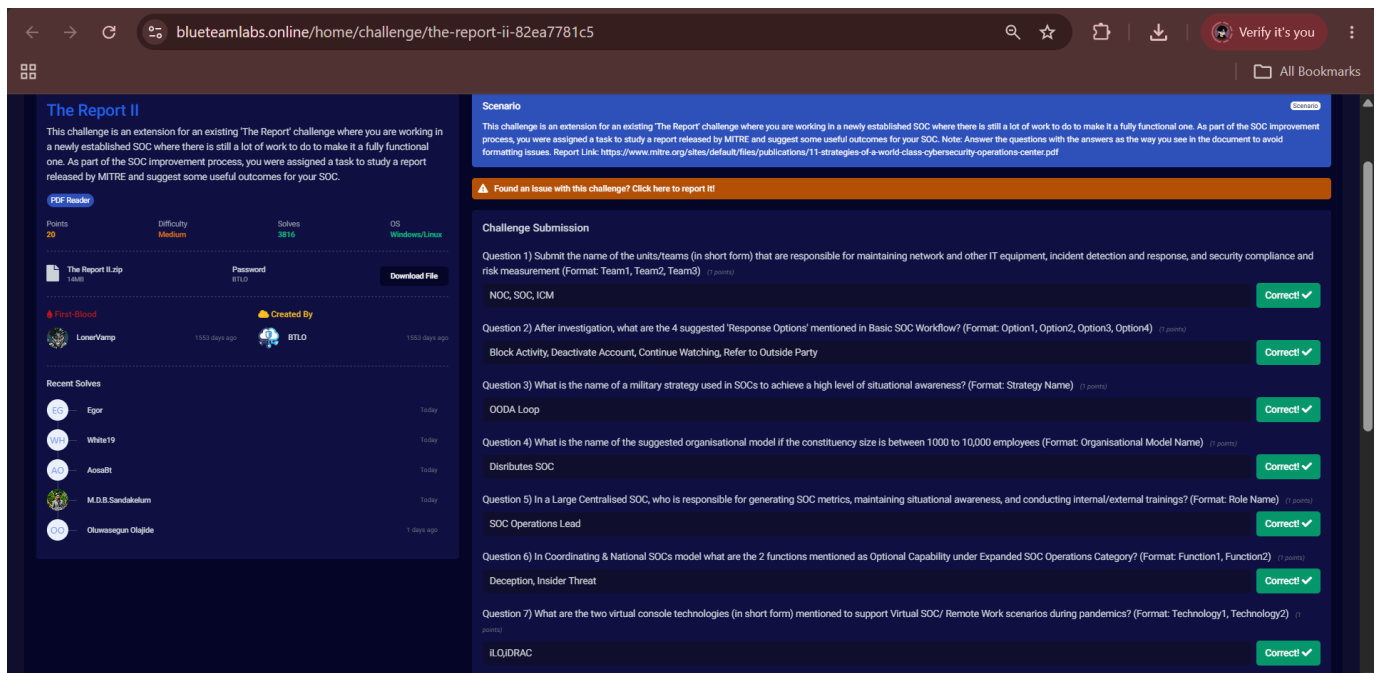
Question 2) After investigation, what are the 4 suggested 'Response Options' mentioned in Basic SOC Workflow? (Format: Option1, Option2, Option3, Option4) (1 points)

Format: Option1, Option2, Option3, Option4 [Submit](#)

32°C Mostly sunny Search ENG UK 03:44 25/09/2026

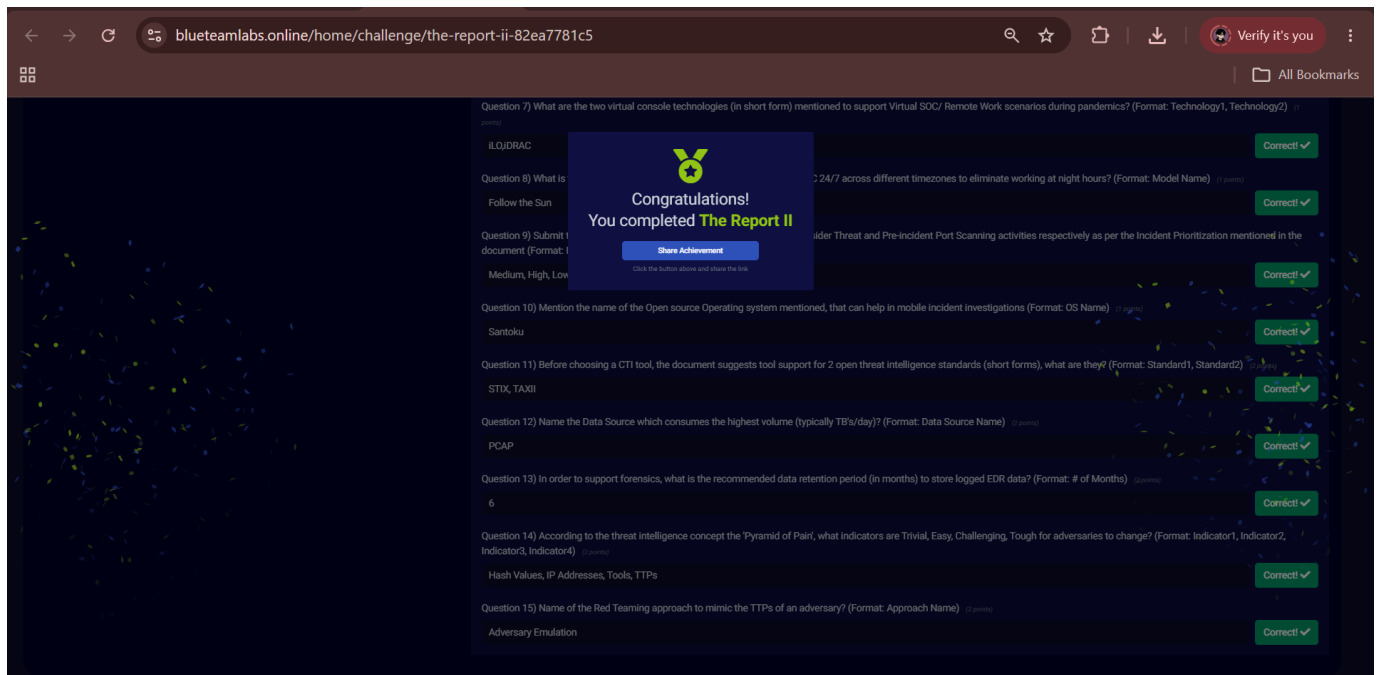
Caption: Blue Team Labs Online shows the selected challenge, The Report II, and its SOC improvement scenario.

Exhibit 7 — BTLO completed answers



Caption: The challenge interface marks the submitted answers as correct, including NOC/SOC/ICM, the response options, OODA Loop, distributed SOC, SOC Operations Lead, Deception/Insider Threat, and iLO/iDRAC.

Exhibit 8 — BTLO completion evidence



Caption: The supplied BTLO completion screenshot confirms the student's challenge result in the platform context.

8. Assessment Questions

1. What is the difference between a raw event, an alert, and an incident? A raw event is an individual observation, such as a file creation or failed login. An alert is a detection rule's notification that one or more events deserve review. An incident is a confirmed or sufficiently credible security situation that requires coordinated response.

2. Why are timestamps important when correlating logs? Timestamps place events in sequence and help the analyst identify what happened before and after an alert. They also reveal whether records from different sources refer to the same activity. Time-zone differences and clock drift must be considered.

3. Why can one log entry be misleading without surrounding events? A single entry may be caused by an approved task, a scanner, a shared system, or a normal user action. The surrounding account, host, process, network, and authentication activity provides the context needed to distinguish normal behavior from attack activity.

4. What is a false positive? Give a SOC example. A false positive is an alert that resembles suspicious behavior but is shown to be benign after investigation. In this lab, the Unusual VPN Login Location alert was marked False Positive after review.

5. What is the purpose of a SIEM? A SIEM centralizes and normalizes security-relevant logs, applies detection rules, supports correlation, and gives analysts a place to investigate alerts and document decisions.

6. When should an L1 analyst escalate instead of closing an alert? Escalation is appropriate when malicious activity is likely, when execution or compromise is confirmed, when a privileged asset is involved, when there is possible lateral movement or data loss, or when the evidence is insufficient for a safe closure and the potential impact is significant.

7. What evidence should be collected before recommending containment? The analyst should collect the alert record, timestamps, affected host and user, process tree, file path and hash, authentication events, DNS and network connections, proxy or browser history, related endpoint detections, persistence indicators, and evidence of execution or lateral movement.

9. Conclusion

This Week 7 assessment demonstrates practical progress from basic SOC concepts to evidence-based alert investigation. The work shows that triage is not simply a process of opening the highest-severity alert. It requires prioritisation, context, correlation, clear separation of event and incident, and a documented verdict.

The primary case study, Double-Extension File Creation, was correctly classified as a High-severity True Positive in the training environment. The next professional step would be to validate execution, scope the affected endpoint, search the hash and domain across the

environment, and escalate when the evidence indicates a broader incident. This approach keeps the investigation concise, defensible, and aligned with L1 SOC responsibilities.

Appendix A — Submission Checklist

Requirement	Status
SOC Fundamentals completed	Included; completion screenshot shows 7 tasks and 128 points
SOC L1 Alert Triage completed	Included; completion screenshot shows 6 tasks and 80 points
One free BTLO defensive challenge completed	Included; The Report II evidence is attached
Own screenshots and date/time evidence	Included; system date/time is visible in supplied screenshots
One investigated alert	Included; Double-Extension File Creation
6+ event incident timeline	Included; six documented timeline rows
5 Ws analysis	Included
Professional SOC investigation report	Included
GitHub repository link	[Insert link before submission]
LinkedIn post link	[Insert link before submission]

References

Prepared for academic submission by Muahmmad Umer Imran (AQT-1170).

Replace the two bold link placeholders on the cover page and in the checklist after publishing the GitHub repository and LinkedIn post.